# ON MEANING AND MANTRAS
## ESSAYS IN HONOR OF FRITS STAAL

Frits Staal at the 2011 *agnicayana* in Kerala, India.
Photos courtesy of Michael Witzel.

# On Meaning and Mantras
## Essays in Honor of Frits Staal

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# Contents

# The *Bhūtasaṃkhyā* Notation: Numbers, Culture, and Language in Sanskrit Mathematical Literature

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## Abstract

A method of expressing numbers by means of symbolic words arranged as in place value notation was developed in India in the early centuries of the Common Era. In the *bhūtasaṃkhyā* notation, the digits from 0 to 9 are denoted by certain significant words that connote a numerical association derived from many areas of Indian culture. This paper analyzes this system, taking into account the relation between mathematical representations, culture, and language that contribute to its originality. While looking at the *bhūtasaṃkhyā* as the creative attainment of a process in which the flexibility of the Sanskrit language has been used as a tool that functions in poetic structure and the method of oral transmission of Sanskrit mathematical texts, I argue that it also represents a "constitutively social form" of language, where each sign, embedded in social images, memories, and descriptive information, becomes mathematical "sense-making," insofar it is rooted in the semantic field "Indian culture."

In order to understand cognitive and social functions served by this numerical notation system, I attempt to delineate, with the use of a formalized language typical of mathematical logic, the dialectic between numbers and collective representations in the context of ancient Indian culture.

## Introduction

Frits Staal has challenged many dominant ideas about language, logic, ritual geometry, and religious practices with implications far beyond the field of Sanskrit.[1] Among his most influential contributions to Indological studies have been his theories on the grammatical method in

Pāṇini. Explaining the different directions taken by scientific developments in India and in the West, Staal writes that "Pāṇini's method has occupied a place comparable to that held by Euclid's method in Western thought."[2] Comparing the methods employed by the Indian linguist (fourth century B.C.E.) and the Greek mathematician (third century B.C.E.), and defending the thesis that the mathematical method is characteristic of Western philosophy while the grammatical method is characteristic of Indian philosophy, Staal emphasizes that

> mathematics and linguistics share certain features that also occur in other sciences but which are especially well-developed in these two: the expressions are fully explicit, and the system as a whole is as simple as possible. This results in the postulation of a small number of elements which are combined in accordance with rules of various types. In India, these requirements led to the development of the sūtra style, which emphasized rules, and which, having originated in the ritual manuals, culminated in Pāṇini's grammar.[3]

In the context of such an analysis, the relation between mathematics and natural language can be approached from different perspectives. For instance, one can emphasize the language-like attributes of mathematics, pointing out that mathematics is indeed itself a language: an artificial language especially designed for carrying out mathematical activity employing a standard code of conventional symbols to express relations, arguments, and sentences. Moreover, the logic, clarity, and set of formulas that regulate the high level of formality of mathematics similarly evoke the rules and principles governing the syntax of a language. It is interesting to note that research on modern linguistics, computer science, and mathematical logic investigates the formal structure of language using mathematical methods.

Certainly, the development of specific linguistic practices in Sanskrit mathematical texts, and consequently the practice of mathematics, must be situated in relation to the cultural context in which they took shape. I argue that the various forms of expression attested in Sanskrit mathematical texts should be considered in the light of the relation between the oral tradition and written literature in ancient India, which contributes to the originality of these texts.

In the attempt to investigate certain properties of the *bhūtasaṃkhyā* system of notation, an ingenious linguistic code through which the versatility of the Sanskrit language enriched the way of communicating mathematical concepts in the Indian tradition, this paper explores the relation between culture, language, and numbers in Sanskrit mathematical texts. For this purpose, I address the following key questions: How has Sanskrit been adapted to the needs of formal scientific expression? What can we say about the way in which mathematical representations and language, two well-defined meaning-making systems, have interacted in construing mathematical knowledge in the Indian context?

It is not my purpose here to demonstrate how the richness and flexibility of the Sanskrit language have enriched the scientific terminology of mathematics. I have shown elsewhere[4] that the linguistic attainment in this area is fascinating, as it shows not only a wide range of technical terms for denoting concepts and their explanations, but also a variety of ways to convey mathematical meanings.

In order to investigate the role played by the bhūtasaṃkhyā in Sanskrit mathematical treatises, in this article I will particularly refer to the *Gaṇita sāra saṃgraha* by Mahāvīrācārya (ninth century C.E.), which I have selected to serve as the main source for certain theoretical and practical considerations.[5] After delineating a brief history of the *bhūtasaṃkhyā,* I will survey the various methods of expressing numerical values found in this text, and then look more closely at the exceptional list of word-numerals[6] given by the author. In exploring the relation between worldviews, numerical values, and semantic fields that may have contributed to formalize the *bhūtasaṃkhyā* system, I attempt to represent its logical structures by means of the symbolic language typically adopted in modern mathematical linguistics and logic. Though my interest primarily concerns the role played by literary and oral traditions in shaping mathematical writings in all their dimensions, I adopt a perspective that takes into account, within the framework of the Indian context and more particularly of the śāstric branches of learning, the influence of social and cultural factors in the representations of mathematical objects.

## Early Occurrences of the Word-numerals System

In order to first investigate the *bhūtasaṃkhyā* as a system of notation and then as a language, I will start by clarifying what a numerical notation system is. In his historical-anthropological analysis, Stephen Chrisomalis defines it as a "visual, relatively permanent, and primarily non-phonetic system for representing numbers."[7] The primary function of a numerical notation as a representational system is to communicate mathematical values. In this regard, I wish to point out that the *bhūtasaṃkhyā,* although it is a structured system for representing numbers based on alphasyllabic numerals, was not designed with mathematical purposes in mind, even though it was mainly employed in mathematical and astronomical works. I will clarify this point in the following sections of this paper.

Before going further with the investigation of this system of notation, let me briefly lay out its historical account. It has been noted by several scholars[8] that by the first half of the first millennium C.E. Sanskrit mathematical and astronomical works had employed mainly two kinds of notation for expressing large numbers consisting of several digits. In the first type of notation, commonly known as *bhūtasaṃkhyā,*[9] the digits from 0 to 9 are expressed as words, while in the second method, called *kaṭapayādi,* they are represented by consonants of the Sanskrit alphabet. Among the two, the *bhūtasaṃkhyā* notation system seems to be older and was employed throughout India from the early centuries of the Common Era on. The first traces of the *bhūtasaṃkhyā* system of numeration, in which numbers are expressed by means of words arranged as in place value notation but following a left-to-right direction,[10] are found in the *Yavanajātaka* by Sphujidhvaja,[11] a third-century C.E. Sanskrit text consisting of an earlier translation into Sanskrit of a Greek astrological text. The emergence of Greco-Indian astrology is part of the cultural fusion that followed the collapse of Alexander's empire at the end of the fourth century B.C.E. During this time, Indian Greeks (called Yavanas in Sanskrit) integrated into Indian culture, adopting Hindu beliefs, taking Sanskrit names, and intermarrying into Indian dynastic families.

Christopher Minkowsky[12] underlines that a new period of Indian astronomy famously began in about the second century C.E. with the appropriation of Greco-Babylonian and Hellenistic astronomical-astrological models, parameters, and interpretative systems. Given this background, the Yavanajātaka represents an Indianized version of traditional Greek astrology, including the first-known occurrence in India of the familiar twelve signs of the Greco-Babylonian zodiac. The *bhūtasaṃkhyā* appears for the first time in this text as a peculiar Indian invention. The function of this verbal notation is to provide synonyms for ordinary words denoting cardinal numbers, such as *eka,* "one," *dvi,* "two," and so on. This system consents to the name of any object or being, associated in nature or by culture with a particular number, to stand for it. In this way, the mathematical referent is expressed without ruining the scansion of the verse.[13]

The *bhūtasaṃkhyā* system was popularly employed in Sanskrit *jyotiḥśāstra* literature[14] as well as in metrics. In this system, the symbolic words that denote numbers, arranged from unit's place onward in a right-to-left sequence,[15] are derived from many sources, including mythology, ritual, the human body, cosmology, poetic convention, and various *śāstra*s. We can infer that this notation came into use for practical reasons: it preserves the rhythm of the *śloka*s, avoids inelegant ways of naming numbers, and supports a mnemonic function, linking words with numbers. Such use of word-numerals is present in almost all later mathematical and astronomical works.

In regard to the second method of notation mentioned above, according to S. R. Sarma[16] the first datable occurrence of the *kaṭapayādi* alphabetic notation is in Haridatta's *Grahacārinabandhana,* composed in 683 C.E. Bhibhutibhisan Datta and Avadesh N. Singh[17] place its invention around 500 C.E. and claim that it must have been known to the mathematical-astronomer Āryabhaṭa (ca. late fifth–early sixth centuries C.E.), but there is no textual evidence to support this assertion. The name *kaṭapayādi* is taken from four syllables of the Sanskrit alphabet (*ka, ṭa, pa, ya*) that are assigned to the digit 1 in this system. Vowels have no numerical significance in the *kaṭapayādi* convention, and neither do consonants that appear in conjunction with a following consonant or at the end of a word. Only a consonant that is immediately followed by a vowel has a numerical value. Chrisomalis writes that "structurally this system is an ordinary ciphered-positional and decimal system."[18] Kim Plofker underlines that the

> flexibility of this system means that authors can encode digit sequences in actual Sanskrit words. For example, the word bhavati, "becomes," is decoded as the sequence 4–4–6, implying the number 644.[19]

This way of using code syllables at a symbolic and semantic level ensures that it is always possible to form code words that also carry meaning.

Around 150 years earlier the mathematician-astronomer Āryabhaṭa employed another technique for verbally representing numbers. In his work the Āryabhaṭīya (1.2) he describes an alphanumeric encoding system that depends on the phonetic order of the Sanskrit alphabet, according to which consonants are assigned specific integer values and vowels are used to represent decimal powers. In this way, consonants and vowels are combined into syllables representing numbers. Annette Wilke and Oliver Moebius emphasize, however, that

> in the *kaṭapayādi* system the sensory dimension of the text is emphasized by choosing beautiful words with pleasant associations. This is probably a

major reason why the *kaṭapayādi* system became so popular, while Āryabhaṭa's system died out. It was simpler and aesthetically more satisfying by allowing a greater play with sounds and meanings.[20]

## The *Bhūtasaṃkhyā* System in the *Gaṇita Sāra Saṃgraha*

The *Gaṇita Sāra Saṃgraha* (GSS) by Mahāvīrācārya, a Jaina mathematician, is the first-known independent mathematical treatise in Sanskrit to survive in its entirety.[21] According to historians, Mahāvīrācarya lived at the ninth-century Rastrakuta court in Karnataka, indicated by his praise to King Amoghavarṣa.[22] The study of mathematics was very popular among Jaina scholars, as it represents one of the four *anuyoga*s, "auxiliary sciences," that indirectly enable the practitioner to attain *mokṣa,* "liberation." As suggested by the title, which translates as "compendium on the essentials of mathematics," Mahāvīrā's work is an exposition, a summary of the essentials of *gaṇita.* I have shown elsewhere that Mahāvīra organizes various teachings into a self-consistent whole and provides an unprecedented amount of detail about the organization of mathematical topics.[23]

In the first chapter, *saṃjñādhikāra,* which is dedicated to the treatment of technical terms, Mahāvīra gives in ten *śloka*s a large list of names of certain things used to denote numbers in arithmetical notation. Though a more or less extensive use of words as numerals can be found in other mathematical works,[24] to my knowledge no other Sanskrit mathematical text provides such an exceptional list. Mahāvīra seems to have been the only mathematician who collected and arranged in order words used for representing numbers. He seems to have introduced this topic with the word, *saṃkhyāsaṃjñā,* "technical terms for numbers."[25] It is interesting to note that Mahāvīra uses terms and concepts belonging to Jain philosophical and spiritual thought, not only those of general Sanskrit literature.

In the following ten verses, Mahāvīra lists words standing for numbers:[26]

*śaśī somaś ca candrendū prāleyāṃśū rajanīkaraḥ //*
*śvetaṃ himagu rūpañca mṛgāṅkaś ca kalādharaḥ //*
*dvi dve dvāv ubhau yugalayugmaṃ ca locanaṃ dviyam /*
*dṛṣṭir natrāmbakaṃ dvandvam akṣicakṣur nayaṃ dṛśau //*
*haranetraṃ puraṃ lokaṃ trai [tri] ratnaṃ bhuvanatrayam /*
*guṇe vaḥniḥ śikhī jvalanaḥ pāvakaś ca hutāśanaḥ //*
*ambodhir viśadhir vārdhiḥ pyodhis sāgaro gatiḥ /*
*jaladhir bandhaś ca turvedaḥ kaśāyassalilākaraḥ //*
*iṣur bāṇaṃ śaraṃ śastram bhūtam indriyasāyakam /*

*pañca vratāni viṣayaḥ karaṇīyastantusāyakaḥ /*
*ṛtujīvo raso lekhyā dravyaṃ ca ṣaṭukaṃ kharam //*
*kumāravadanaṃ varṇaṃ śilīmukhapadāni ca //*
*śailamadrirbhayaṃ*[27] *bhūdhro nagācalamunirgiriḥ (sic) /*
*aśvāśvipannagā*[28] *dvīpaṃ dhāturvyasanamātṛkam (sic) /*
*aṣṭau tanurgajaḥ karma vasu vāraṇapuṣkarim //*
*dviradaṃ dantī digduritaṃ nāgānīkaṃ karī yathā /*
*nava nandaṃ ca randhraṃ ca padārthaṃ labdhakeśavau //*
*nidhiratnaṃ grahāṇāṃ ca durganām ca saṅkhyayā /*
*ākāśaṃ gaganaṃ śūnyam ambaraṃ khaṃ nabho viyat //*
*anantam antarikṣaṃ ca viṣṇupādaṃ divi smaret //*[29] (GSS 1.53–63)

Based on these verses, I present the following correspondence between numbers and words:[30]

(1) *Śaśin, soma, candra, indu, prāleyāṃśu, rajanīkaraḥ, himagu, mṛgāṅka,* and *kalādharaḥ* are synonyms of "moon," which is only one; *śveta,* lit., "white," stands also for "moon," and *rūpa,* "form/shape," denotes the number 1 because everything has only its own one shape.

(2) *Yugala, yugma, dvi,* and *dvaṃdva* mean "pair, couple," and therefore stand for the number 2, as well as *locana, dṛṣi, dṛṣṭi, netra, ambaka, akṣi,* and *cakṣu,* which are synonyms of "eyes." *Naya* means "method"; according to the Jainas there are two methods of knowledge (*pramāṇa* and *naya*).

(3) Haranetra refers to the three eyes of Śiva; *pura,* "city," is linked to the three cities representing three *asura*s, which, as found in the *Purāṇa*s, are said to have caused great distress to the gods until Śiva destroyed them. *Triratna,* "three jewels," refers to the three jewels mentioned in Jainism: right perception, right knowledge, and right conduct. *Loka* and *bhuvana* mean "world"; the number of worlds ordinarily counted in Sanskrit literature are three: the upper, lower, and middle worlds. *Guṇa* denotes the three qualities of *sattva, rajas,* and *tamas. Vahni, agni, jvalana, pāvaka,* and *hutāśana* are synonyms for "fire"; there are three kinds of sacrificial fires: *gārhapatya, āhavanīya,* and *dakṣiṇa.*

(4) The words *ambodhi, viśadhi, vārdhi, payodhi, sāgara, jaladhi,* and *salilākaraḥ* mean "ocean." It was believed that there are four oceans, the eastern, southern, northern, and western oceans. The word *gati,* "passage," refers to the "passage into rebirth"; according to Jainism the soul may have four kinds of embodiment (*deva, tiryak, manuṣya,* and *naraka*). *Bandha* is "bondage," of which the Jainas recognize four kinds of spiritual bondage (*prakṛti, sthiti, anubhāva,* and *pradeśa*). The word *kaṣāya,* "attachment to objects," is also connected to Jaina philosophy; there are

four causes of attachment (*krodha, māna, māyā,* and *lobha*). Last, the word *veda* stands for the number 4, referring to the four Vedas.

(5) *Iṣu, bāṇa, śastra, śara,* and *sāyaka* are synonyms for "arrow"; the arrows of the Indian "Cupid" (Kāmadeva) are said to be five. *Bhūta* is one of the five elements; *indriya* denotes one of the five senses; *karaṇīya* and *vrata* refer to the five acts of devotion according to the Jaina religion; and *viṣaya* alludes to the objects cognizable by the five senses.

(6) According to Sanskrit literature there are six seasons (*ṛtu,* "season") and six principal tastes (*rasa*). Dravya concerns the elementary substances, which are six (*jīva, dharma, adharma, padula, kāla,* and *ākāśa*) according to the Jainas. Kumāravadana refers to the six faces of Kumāra (Śiva's son). Śilīmukhapada literally means "leg of a bee," of which there are six; thus the word stands for the number 6.

(7) *Śaila,* as well as *acala, mahīdra, bhūdhra, naga,* and *giri,* are synonyms for "mountain." The geography of the *Pūrana*s recognizes seven principal mountains. *Muni* stands for the seven luminous and eternal sages mentioned in Sanskrit literature. *Aśva* alludes to the seven horses of the sun's chariot; *aśvin* literally means "consisting of horses." *Pannaga* are the seven serpents reckoned in Hindu mythology. *Mātṛkā* stands for the seven goddesses. *Dhātu* refers to the seven constituents of the body.

(8) *Ibha, gajaḥ, vāraṇa, puṣkarī, dvirada, dantī, naga,* and *karī* are synonyms for "elephant"; eight elephants are said to guard the eight cardinal points of the world. *Tanu* means "body," and stands for the number 8, as Śiva's body is considered to be made up of eight elements. There are also eight kinds of karma, according to the Jainas. *Vasu* alludes to the eight classes of Vedic deities. *Dik* means "direction" and stands for the eight cardinal points. *Durita* means "bad action," of which there are eight types, just as there are eight kinds of *karman* ("action"). *Anīka* refers to the eight kinds of armies mentioned in Sanskrit literature.

(9) *Nanda* is the name of a dynasty of nine kings that ruled in Magadha. *Randhra,* "cavity, hole," refers to the nine orifices of the human body. *Padārtha* means "objects"; the Jainas recognize nine categories of things. *Labdha,* "obtained," concerns the attainment of the nine powers recognized in Jaina philosophy. *Keśava,*[31] *nidhi,* and *ratna* are synonyms of "jewel,"[32] while *graha,* "planet," alludes to the nine planets recognized by Indian astronomy.

(10) *Ākāśa, ananta, gagana, ambara, kha, nabha, viyat, antarikṣa, viṣṇupāda, śūnya,* and *dyu* are all synonyms for "sky" (or "space") and stand for the number 0.

The information provided by this list represents a practical device given by Mahāvīra for interpreting the coding-based association of the word-numerals system on which some of his *uddeśaka*s, "sample problems," are further expressed. As he does not use all the words mentioned, the list may have been intended as a technical tool for mathematical students trained in *jyotiṣa* literature.

Let us now look more closely at the way in which the *bhūtasaṃkhyā* system is applied in the GSS. Though actually well distributed throughout the work, it is more commonly used in the second chapter (*parikarmavyavahāra*), devoted to the explanation of the arithmetical operations. In verses 10–17, for instance, certain numbers are said to constitute different families of "necklaces," referring to the arrangement of the figures.

Mahāvīra gives the following *uddeśaka* in *śloka* 13, using the *bhūtasaṃkhyā* system:

> *analābdhihimagumuniśaraduritākṣipayodhisomamāsthāpya* /
> *śailena tu guṇayitvā kathaya tvaṃ rājakaṇṭhikābharaṇam* // (GSS 2.13)
>
> Having written down: 1 (*somam*, "moon"); 4 (*payodhi*, "ocean"); 2 (*akṣi*, "eyes"); 8 (*durita*, "wordly action"); 5 (*śara*, five "arrows"); 7 (*muni*, seven "sages"); 1 (*himagu*, "moon"); 4 (*abdhi*, "ocean"); 3 (*anala*, three sacrifical "fires"); [and] having multiplied [the number so obtained] by 7 (*śailena*), give [the result that is like] a royal necklace.

Therefore, the two numbers expressed by the *bhūtasaṃkhyā* method are 142857143 and 7 respectively, which when multiplied yield "the royal necklace," 1000000001.

## Numeral Expressions in the GSS

We have seen so far that the *bhūtasaṃkhyā* is a particular way to express numerical values using a word-numerals association, but there are other methods in which Sanskrit was applied for expressing numbers. For the sake of metrical convenience as well as for aesthetic-expressive needs, various devices are used in Sanskrit mathematical literature. In regard to the GSS one finds, for example:

The additive method, one of the most common in mathematical works:[33]

> *tryadhikadaśatriśatayutāny ekatriṃśatsahasrajambūni* /
> *saikāśītiśatena prahṛtāni narair vadaikāṃśam* // GSS 2.25
>
> 31313 *jambū* fruits have been divided between 181 persons. Give the portion of each.

The number 31313 is here given as (reading from right to left) *ekatriṃśat sahasra* = 31,000 added to (*yutāni*) *try adhika daśatriśata* = 3 added to (*adhika*, lit., "additional") 310 = 313. The number 181 is expressed as *aśītiśata,* which is 180 preceded by *saika* (lit., "with 1"), therefore 181.

The subtractive method is also often used. For instance, in GSS 2.4 the number 139 is expressed as *catvariṃśac caikonaśatādhika*, literally, "100 (*śata*) defective (*ūna*) of 1 added to (*adhika*) 40 (*catvariṃśat*)," therefore (100 - 1) + 40.

The multiplicative system is another method used by Mahāvīra. For instance, the expression *pañcatriṃśac chatasamadhikaṃ saptanighnaṃ sahasram* (GSS 2.35) stands for the number 7,135, given as "1,000 multiplied by 7 (*saptanighnaṃ sahasram*) with the addition (*samadhikaṃ*) of 135 (*pañcatriṃśat śata*)," therefore (1,000 x 7) + 135.

In *śloka* 27 Mahāvīra uses a rather interesting expression for the number 12345654321:

> *ekādiṣaḍantāni krameṇa hīnāni hāṭakāni*
>
> [A number of] pieces of gold [such as] beginning with 1 and ending with 6, and then diminishing in order.

Verse 15 belongs to the *uddeśaka*s mentioned above, in which certain numbers are said to constitute different families of "necklaces":

> *sapta śūnyaṃ dvayaṃ dvandvaṃ pañcaikañ ca pratiṣṭhitam /*
> *trayaḥ saptatisaṇgunyaṃ kaṇṭābharaṇam ādiśet* // 15
>
> The figures 7, 0, 2, 2, 5, and 1 are put down [in order from the units' place] (reading from right to left), then this number, which is multiplied by 73, should be called a necklace (*kaṇṭābharaṇam*).

The product of 15,2207 × 73 is 11,111,111.

It is interesting to note that the first three methods are place value–notation based, meaning that the numbers are expressed accordingly to the their position in the decimal system. By contrast, in the fifth method a certain large number is given as a list of digits, one after one and read from right to left.

If, on the one hand, the variety of these systems represents a particularly elegant way of playing with the beauty of numbers, on the other it clearly allows for dispensing with flexible systems for writing numerical expressions while respecting the literal requirements of the verse.

In light of this, it would take us much too far afield to go into detail about the lexicon and etymology of technical terms used in Sanskrit mathematical texts. For my purpose here, it is enough to highlight that the vocabulary used, typical of an ordinary, "natural" Sanskrit, denotes a wide range of conventional terms that have been fully integrated into mathematical context, with the consequent acquisition of technical meanings. As I have illustrated elsewhere,[34] the result is an established "specialized lexicon," with some of these terms being consistently polysemic.

## Means of Knowledge: The "Eight Qualities of a Mathematician"

In focusing attention on the linguistic practices of Sanskrit mathematical texts, I want to suggest that the various systems of writing numerical expressions found in mathematical texts should also be considered as purely functional literary devices to the specific features of Sanskrit literature, where the influence and privileged position of oral transmission is evident even in treatises on technical subjects, often considered as knowledge to be learned by heart. For instance, metrical form and brevity render the mathematical text compact and therefore more easily memorized; at the same time, while reading a Sanskrit mathematical text we can notice that many details are missing in the concise verse-rules, so much that sometimes such texts are difficult to understand. They were probably orally transmitted, and the reasoning beyond mathematical statements were also orally expounded by teachers.

However, it is important to keep in mind that Sanskrit mathematical treatises were often accompanied by long prose commentaries. The role of commentaries[35] was not only to explain the literal meaning of verses but also to convince the reader that they were properly formulated. Commentators usually repeat the compact verse rule in a more articulated and comprehensible prose form, giving sample problems and explaining in more or less detail, according to each author's style, the mathematical procedures, often step by step. The pedagogical goal of a commentary, in representing the perspective of the *mūlākāra,* the "author of the root-text," commented upon, is to guide the audience's reading in such a way that the reader's understanding of the mathematical content is improved.

Another significant aspect is that the validity of the statements are not discussed in Sanskrit mathematical commentaries, because the authority of the śāstric traditional knowledge is considered to be the source and indirectly establishes the texts' truth and accuracy.

In this regard, the role of the mathematical proof in Indian mathematics is a controversial issue among scholars. Although historiography has not paid adequate attention to the role played by the mathematical examples in the Indian tradition, I have shown elsewhere[36] that *uddeśaka*[37] does not presuppose merely the practical execution of an exercise, but rather represents a significant device enabling the construction of an actual proof, and therefore provides a system of knowledge based on deductive syllogism. I argue that in Sanskrit mathematical texts "the sample problem" highlights the relation between the type of text and the kind of proof developed. In my view, sample problems in Sanskrit mathematical texts enable authors to formulate general proofs to mathematical problems.

This point relates closely to the epistemological foundations of ancient Indian mathematics. Textual sources do not help provide a detailed account of the way in which mathematical knowledge was perceived and acquired and, as Plofker has highlighted,[38] there is a great deal of uncertainty concerning how mathematical practitioners were trained and evaluated.

For these reasons, the following passage from GSS is extremely valuable and of great interest. Mahāvīra lists here what he calls the eight "qualities of a mathematician," and in so doing he expresses the view of the tradition of which he was a part on the "means of knowledge" for a mathematician:

> *laghukaraṇohāpohānālasyagrahaṇadhāraṇopāyaiḥ /*
> *vyaktikarāṅkaviśiṣṭair gaṇako' ṣṭābhir guṇair jñeyaḥ* // GSS 1.69
>
> A mathematician is to be known by eight qualities, which are paramount for the manifestation of numbers: concision, [a thorough knowledge of mathematical] procedures, confutation, not indolence, comprehension, memory, and by [ingenious] means of working.

Before looking at the qualities mentioned here, I must point out a problem concerning this verse.[39] I understand *vyaktikarāṅkaviśiṣṭair* to qualify *aṣṭābhir guṇair,* but in Raṅgācārya's translation this represents the eighth and last *guṇa,* "quick method in working."[40] If *laghukaraṇa* is a compound, however, we would find *laghū* instead of *laghu,* since when compounded with forms of the verb *kṛ* the first member (either noun or adjective) ending in a vowel lengthens that vowel.[41] A similar issue occurs in the second line with *vyaktikara,*[42] which should actually be *vyaktīkara.*[43] In my opinion, it is only the first line to list the eight attributes of a mathematician. I thus consider *laghu* as standing for *laghutva* and *karaṇa* as denoting the second *guṇa.*[44]

Having pointed out the ambiguity of this verse and in order to understand to what extent they pertain to the specific activity of a mathematician, let us look more closely at each of these qualities, or *guṇa.* The first *guṇa* is *laghutva,* which indicates the quality of concision, economy expressed as the ability to go straight to the point. The second quality is *karaṇa,* which in Sanskrit mathematical texts means "procedure, method, calculation." It refers here to the algorithmic ability and knowledge a mathematician must possess.

The third quality, *ūha,* "inference,"[45] suggests the idea of a cognitive act, a process of deducing a conclusion from evidence and reasoning, thus "inductive reasoning." This implies a sense of cause and effect, the ability to construct and follow a step-by-step logical argument. Much of mathematics depends on "if this, then that" reasoning, an abstract form of thinking about causes and their effects.

The fourth is *apoha,* lit., "exclusion,"[46] which I understand here as alluding to the ability to remove doubts and confute others' ideas, and therefore refers to the way a mathematician should deal with objections. The fifth is *anālasya,* "not indolence." The sixth is *grahaṇa,* "comprehension": a mathematician should be sharp in learning. The seventh is *dhāraṇā,* "memory." The eighth and last is "[ingenious] means of working," the ability to find solutions, to construct and follow a causal chain of facts or events.

All of these can be recognized as general qualities of intellectual cognition and at the same time, as the author seems to suggest, they should represent the characteristics and the methodological setting of a mathematician. Surprisingly, I have found four of these "*guṇa*s " mentioned in *Arthaśāstra* 6.1.4,[47] which in turn seems to run parallel to the *aṣṭabuddhiguṇa*s, "eight qualities of the intellect," mentioned in Purāṇic texts.[48]

A final consideration concerns the number 8, which is often found in Jaina literature.[49] For instance, according to Jain texts there are eight kinds of fears, eight main types of karma, eight *vargaṇa*s,[50] eight virtues, eight objects of auspiciousness, and eight items used to perform the *pujā* of a *tirthaṅkara* in a Jaina temple.[51]

## Culture and Mathematical Representations: The *Bhūtasaṃkhyā* as a Language

A major factor in investigating the *bhūtasaṃkhyā* is to understand the logic process that contributed to the development of this notation system. It can scarcely be ignored that recent research in the field of cognitive psychology would certainly provide an important perspective in understanding processes such as memory, perception, mental imagery, association, concept formation, problem solving, and pattern recognition that influence the elaboration of symbolic representations (such as the *bhūtasaṃkhyā*) in systems of knowledge. In his work *Numerical Notation: A Comparative History,* Stephen Chrisomalis repeatedly points out that the social dimension of numerical notations is intimately linked to a specific historical context, and at the same time implies cognitive factors.

The intriguing *modus operandi* of the *bhūtasaṃkhyā* evokes the relation between language, mathematics, and reality. My arguments in the following discussion draw on principles of cognitive psychology (How does the brain processes information and concept formation?), the philosophy of mathematics (What is a number and what is a mathematical object? How and to what extent mathematical ideas are shaped by cultural and cognitive features?), and the sociology of scientific knowledge (What are the social conditions that affect science?), and I often make use of the formalized mathematical language typical of Western logic and linguistics.

As it is a method in which conventional signs of the language-system "Sanskrit" are used to describe another system of signs, namely the mathematical language of numbers, I propose to consider the *bhūtasaṃkhyā* as a meta-language. The question here is to identify what kind of act is implied when semantically different terms are used as referents for the same numerical value. Adopting the well-known Sausserian categories,[52] we can say, for instance, that in the *bhūtasaṃkhyā* language terms such as *abdhi,* "ocean," and *gati,* "passage," become signifiers of the same signified number 4.[53]

Now, according to what principle did linguistic units that denote two different concepts, such as "ocean" and "passage," develop into mathematical objects? We have seen that the word *agni,* "fire," becomes a symbol of the number 3 because it is associated to the idea of the (Vedic) "three sacrificial fires."

Central to a theory of the *bhūtasaṃkhyā* system as a meta-language is the understanding of the mechanism, which we can observe is based on a sort of "affinity by abstraction," that links concepts to corresponding numerical values. As I mentioned above, while looking at the *bhūtasaṃkhyā* as a representational system of communication the first impulse is to emphasize the so-called "cognitive aspects of meaning," those aspects that go to determine the objects spoken of, and what properties they are said to have. This finds an echo in Donald Davidson's much quoted statement: "What is it for words to mean what they do?"[54] Modern philosophers of language, such as Sara Mills, emphasize that

> meaning is constructed by the system of representation and representation is an essential part of the process by which meaning is produced and exchanged between members of a culture. It does involve the use of language, of signs and images which stand for or represent things. Representation connects meaning and language to culture.[55]

I am not attempting here to apply modern semantic studies to define the way meanings are conveyed through signs in the context of literary Sanskrit practices. Rather, the point I want to put forward is that the *bhūtasaṃkhyā,* which is first of all the result of a poetic-literary strategy in which the referent (a certain number) is denoted by a representational signifier (a word taken from everyday usage), evolved as a social convention through the common practice of a group of initiates.[56] In order to outline my arguments more precisely and to illustrate how we can analyze the logical structure of this system, I will introduce "the use of a mathematical language inspired by the language of contemporary logic,"[57] which helps avoid the ambiguity of the natural language and simplifies the dialectic relation of the constitutive elements of the word-numerals system of notation. If objections such as "Have Indian structures been forced into Western categories? Have modern mathematical logic and a Western conceptual framework been imposed onto the Indian tradition?" are raised, I answer with Staal's words:

> I am not preaching an absolute, unchanging truth; my attempt was a scientific inquiry, hence not final. I was careful not to introduce concepts of Western logic into my language by symbols that stand in one-to-one correspondence to the original expressions. . . . I use symbols merely because they are more precise and unambiguous than ordinary English.[58]

Just as Staal's *Universals: Studies in Indian Logic and Linguistics* illustrates his arguments on language and logic in Indian thought, I believe that the formal symbolic representations of mathematical logic can help in the investigation of the principles of the *bhūtasaṃkhyā.* Moreover, it must be acknowledged that the level of systematization, formalization, and conceptualization that characterize modern mathematics and mathematical linguistics on the one hand, and the structure of the language as well as the methods of Indian logic on the other, draw attention to the relation between mathematics, language, and philosophy.

## Number, Object, Quality, and Technical Term: The Four Logical Terms

In the following section I articulate a logical argument consisting of four terms:

(1) *Saṃkhyā,* "number," is in the *bhūtasaṃkhyā* notation that which needs to "be signified" by another element standing for it.

(2) I use the word *bhūta* to denote an animate or inanimate object that, before being adopted as a technical term, is semantically related to the referent conveyed by its literal meaning.

(3) With *saṃjñā* I indicate a "technical term," which I understand as the result of a process by which an object (*bhūta*) has been established to stand for a number.

(4) Last, I call *lakṣaṇa,* lit., "quality, attribute," the characteristic of an "object" that is either socially produced by the process of interaction of the members of the society, or as part of an object by nature. For instance, in the *bhūtasaṃkhyā* system the "object," *akṣi,* "eye," stands for the number 2 because the *lakṣaṇa* of *akṣi* is the fact that human beings have two eyes. This is an inherent "attribute" of the "object" *akṣi.* On the other hand, *naya,* "method," stands also for the number 2 based on the *lakṣaṇa* of *naya,* the two methods of knowledge for the Jainas. This is a "quality" that has been socially attached to the "object" *naya.* Also, *lakṣaṇa* as the *conditio sine qua non* of a certain "object" (*bhūta*) becomes a "technical term" (*saṃjñā*). In a few words, then, when an "object" has such a "quality," the "object" becomes somehow "measurable," and is therefore transformed into a "technical term" (*saṃjñā*) that denotes a "number" (*saṃkhyā*).

Now, let us apply these four logical terms to the word *agni,* which is used in the *bhūtasaṃkhyā* system to stand for the number 3:

(1) The number 3 is the *saṃkhyā.*

(2) *Agni* in its literal meaning of the object "fire" is *bhūta.*

(3) *Lakṣaṇa* is the measurable quality, "three sacrificial fires."

(4) Because the "object" fire bears a *lakṣaṇa* in relation to the number 3, *agni,* "fire," therefore becomes a *saṃjñā* standing for that number.[59]

I suggest that the first, fundamental relation between these terms has the following form:[60]

(1) A (x, y)

This may be read as: "x occurs in y," therefore the relation A is an "occurrence relation."

In relation to our logical argument, where l stands for a "quality" (*lakṣaṇa*) and b for an "object" (*bhūta*), the first relation established is:

(1) A (l, b)

which is read as: "l occurs in b," therefore the relation A is established. Such relation is called "occurrence relation."

The second step, in which S is the *saṃjñā* and s is the *saṃkhyā,* is:

$A\ (l, b)\ [(b = S) \wedge (S\ (:=)\ s)]$

which may be read as: "if the relation A occurs, then b is equal to S and S stands for s."

By means of the relations A, a certain "object" (*bhūta*) can be established as a "technical term" (*saṃjñā*) standing for the number, its "quality" is in relation with, and therefore recognized as, a sign of the codified language *bhūtasaṃkhyā.*

Of course, the negation of the occurrence relation A, which represents here the premise, implies the negation of the other two identity relations.

The logical relations between the variables b, l, and s can be also expressed by the following syllogism consisting of two premises of the form R (x, y), "x is in relation to y," from which one conclusion of the same form is derived.[61] Here, *lakṣaṇa* clearly appears to be the middle term linking the two extremities *bhūta* and *saṃkhyā:*

R (b, l)
R (l, s)
R (b, s)

which can be read as: "if b is in relation to l, and l is in relation to s, then b is in relation to s." In other words, if a given "object" (*bhūta*) is in relation to a certain "quality" (*lakṣaṇa*), and this is in relation to a "number" (*saṃkhyā*), then the given object is in relation to that number.

Another important feature that can be noted concerns the fact that in the *bhūtasaṃkhyā* a certain number can be represented by more than one technical term. This condition can be expressed with:

$R\ (n\ (s_b x, s_b y) \Leftrightarrow R\ (n\ (l_b x, l_b y)$

which is read as: "the relation of n with $s_b x$ and with $s_b y$ occurs only and if only the relation of n with $l_b x$ and with $l_b y$ occurs." In this formulation, n denotes a certain number, $s_b x$ is the "technical term" of a certain "object" x and $s_b y$ the "technical term" of a certain "object" y, while $l_b x$ is the "quality" of the "object" x and $l_b y$ is the "quality" of the "object" y. In other words, a certain "number" can be represented by two different "technical terms" only and if only the "qualities" of the two respective "objects" are informed exactly by the same "number."

## The *Bhūtasaṃkhyā* as Literary Practice, Mathematical Language, and Cultural Representation

After having attempted to outline the structure of the *bhūtasaṃkhyā* by means of symbols and mathematical models of modern logic, I would like to come back to the relation between "number," "object," and the (culturally established) "quality" of the object in the sociocultural context of Indian thought, which affected the development of this peculiar system of notation. At this point, my question is: Can we assert that numbers embody worldviews? While this supposition cannot be developed here, due to limitations of space, if we characterize scientific knowledge as a material and social construction, in which ideas and logical concepts are products of and supported by collective practices, we must recognize that mathematical activities also are an echo of social representations.

I would suggest that we consider the *bhūtasaṃkhyā* as a conceptual map used by those (the "codifiers") who share the interpretative process of the relationship between the four logical terms introduced above, "number," "object," "quality," and "technical term." It thus becomes a means of communication only for certain specialists within the Sanskrit learned tradition. As a literary practice, it developed from the needs of adopting Sanskrit for formal scientific expression and further elaborated it in a mechanism that reflects cultural identities. This connotes the *bhūtasaṃkhyā* as a "constitutively social form" of language, since what I defined as the "quality" (*lakṣaṇa*) of an "object" embodies the customs and traditions of Indian culture (Hindu, Buddhist, and Jain).

The list given by Mahāvīra in GSS 1.53–63 (see above) shows clearly that the technical terms used in the *bhūtasaṃkhyā* system are derived from different cultural and social areas. Each sign of this language becomes an "object of reality," a "sense-making" insofar as it is rooted in the semantic field "Indian culture," because it is not meaningful in itself. What I mean is that each sign is a "linguistic act" that cannot be extracted from its specific cultural background, and which cannot be valid without an effective process of interpretation that only the "codifiers" can put into practice. In other words, each sign of the *bhūtasaṃkhyā* is mathematically effective in universal terms. It acquires its own history and connotation that are familiar only to members of the sign-users' culture, who are able to interpret "the code" that governs the relationship between words and numbers by means of established, shared cultural values.

There are many ways in which cultures contribute to the design and shape of their texts. Sanskrit and mathematical texts took shape in working communities that developed conventions for reading and working with them. I do not want to risk imposing the conceptual framework of Western logic, but let me adopt Sara Mills' famous example of traffic lights.[62] According to the constructionist approach, of which Mills is an exponent, colors and the language of traffic lights work as a signifying or representational system. In the language of traffic lights it is the sequence and position of the colors, as well as the colors themselves, that enable them to carry meaning and thus function as effective signs. It is the code that fixes the meaning, not the color itself. So, as with the language of traffic lights, the meanings carried by the word-numerals are relational.

Meanings are produced within history and culture, as Jonathan D. Culler underlines,

> because it is arbitrary, the sign is totally subject to history and the combination at the particular moment of a given signifier and signified is a contingent result of the historical process.[63]

In this regard, Roland Barthes describes two levels of meaning: "denotation," which is the definitional, literal meaning of a sign (for instance, what the dictionary provides), while "connotation" is used to refer to the sociocultural (and personal) associations of the sign.[64] The "denotative" meaning of a sign is broadly agreed upon by members of the same culture. "Connotations" are determined by the codes to which the interpreter has access; cultural codes provide a connotative framework. Applying these categories to the *bhūtasaṃkhyā* language, we can say that the definitive meaning of *agni*, "fire," is linked to its denotative meaning as "three sacrificial fires," so that the association with the number 3 can be established.

In concluding this section, I would like to mention another perspective in relation to my analysis of the *bhūtasaṃkhyā* as a literary practice, a mathematical language, and a cultural representation, given in the so-called "Descriptivism," an approach to modern semantics generally attributed to Gottlob Frege and Bertrand Russell. In this regard, Genoveva Martí[65] provides an account of the mechanism of reference according to the theory of names elaborated by Descriptivists. Of course, while this line of inquiry is interesting this is not the place to investigate such a detailed approach;

what I want to bring forward here is the assumption that if names are semantically associated with descriptions, then by themselves and because of what they mean they find their referents. Descriptivists argue that speakers attach to names not only descriptive information, they also associate images, memories, and all sorts of connotations. All these associations undeniably play a very important role in cognition.

In a similar way, "objects" of everyday life become "technical terms" in the *bhūtasaṃkhyā* language because of the descriptive information, images, memories, and other associated connotations (what I previously defined as *lakṣaṇa*) attached to them by speakers of the network comprised of the Sanskrit branches of learning, and more broadly in Indian culture.

## Conclusion

What are the implications of all of this for our understanding of textual practices in the Sanskrit mathematical tradition? As Pierre-Sylvain Filliozat writes, "[T]he Sanskrit mathematical text is a literary text; it imitates the form and spirit of the poetic text. . . . Sanskrit poetry is always situated in the structure of orality."[66] The mathematical text remains an "oral text," which can be transmitted through memory, used mentally, and is thus well adapted to such functions: propensity to orality, concision, emphasis on the essential point, links from formula to formula, and metaphorical expressions are some of the structural features of Indian mathematical texts. Moreover, I would argue that the shape of a specialized and well-defined lexicon conveys economy and recognizability to the elucidation of mathematical ideas, at the same time serving oral-mnemonic expository techniques as well as the formal structure of textuality.

At this point, my concern can be summarized in the following question: Can one define the *bhūtasaṃkhyā* system as an "artificial language"? In his article on artificial languages, Staal distinguishes between artificial and scientific Sanskrit.[67] In his definition, artificial Sanskrit is "any kind of artificial language intentionally created to deal with scientific problems, but deviating in some important respect or respects from ordinary, natural Sanskrit," while scientific Sanskrit is "any Sanskrit used for the expression of scientific statements or truths." He argues that although a wide range of technical terms are used, the Sanskrit language applied in mathematical texts is ordinary, "natural" language; there is a specialized lexicon and a technical terminology but this does not make the language an "artificial Sanskrit." Staal points out that during the first millennium

Indian mathematicians employed abbreviations and artificial notations,[68] but their Sanskrit was not artificial Sanskrit and they did not use other artificial language.

What I have attempted in this paper is to analyze the logic that characterizes the *bhūtasaṃkhyā* system as a linguistic practice and as a mathematical language. Having explored the interplay of cultural and cognitive aspects that may have influenced the peculiar development of this system, I suggest that the *bhūtasaṃkhyā* is the result of a process by which the flexibility of Sanskrit language has been used as an active tool functional to poetic structure and to the method of the oral transmission of Sanskrit texts. Moreover, as was rightly highlighted by Frits Staal,[69] it clearly shows that mathematical Sanskrit continues to be Sanskrit.

## Notes

1 My thanks to Dr. Hugo David, Dr. Vincenzo Vergiani, Aleesandro Passi, and Marco Franceschini for their helpful comments on this paper.

2 Frits Staal, *Universals: Studies in Indian Logic and Linguistics* (Chicago and London: University of Chicago Press, 1988), p. 158.

3 Staal, *Universals: Studies in Indian Logic and Linguistics,* p. 31.

4 See Alessandra Petrocchi, *The Gaṇita Sāra Saṃgraḥa: A Linguistic Approach to a Sanskrit Mathematical Treatise* (MPhil Thesis in Sanskrit, Cambridge University, 2012).

5 These considerations concern the variety of systems for numerical expressions used by Mahāvīrācārya, his extensive use of the *bhūtasaṃkhyā,* and questions emerging from details given in this text that are found nowhere else.

6 In the modern historiography of mathematics, Bhibhutibhisan Datta and Avadesh Narayan Singh, in their 1935 work *History of Hindu Mathematics: A Source Book* (Bombay: Asia Publishing House, 1962, reprint), pp. 53–69, coined the term "word-numerals," which is generally used to denote the *bhūtasaṃkhyā.*

7 Stephen Chrisomalis, *Numerical Notation: A Comparative History* (Cambridge: Cambridge University Press, 2010), p. 3.

8 See Kim Plofker, *Mathematics in India* (Princeton, NJ: Princeton University Press, 2009); Sreeramula Rajeswara Sarma, "On the Rationale of the Maxim Aṅkānāṃ Vāmato Gatiḥ," *Gaṇita Bhāratī* 31/1–2 (2009): 65–86; Datta and Singh, *History of Hindu Mathematics.*

9 According to Sarma, "On the Rationale of the Maxim Aṅkānāṃ Vāmato Gatiḥ," p. 67, Sundararāja was the first to use the term *bhūtasaṃkhyā,* at the end of the fifteenth century.

10 Datta and Singh, *History of Hindu Mathematics,* pp. 16–88, investigate the history of place value notation in India.

11 See David Pingree, *The Yavanajātaka of Sphujidhvaja,* 2 vols. (Cambridge, MA: Harvard University Press, 1978).

[12] Christopher Z. Minkowsky, "Astronomers and Their Reasons: Working Paper on Jyotiḥśāstra," *Journal of Indian Philosophy* 30/5 (2002): 495–514.

[13] In 1503, Jñānarāja of Pārthapura composed the *Siddhāntasundara.* Plofker, *Mathematics in India,* p. 289, points out that he uses the number system with a literary and mathematical double meaning: if his sample problems are read with the standard numerical interpretations, the numbers provide quantities for solving the problem; but if read with their literal verbal meanings, they transform the problem into little anecdotes with no mathematical values. See the recent translation of Jñanarāja's work in Toke Lindegaard Knudsen, *The Siddhāntasundara of Jñānarāja: An English Translation with Commentary* (Baltimore: Johns Hopkins University Press, 2014).

[14] The term *jyotiṣa* includes the disciplines of mathematics, astronomy, and divination. See Pingree, *The Yavanajātaka of Sphujidhvaja,* vol. 1, p. 198; Plofker, *Mathematics in India,* p. 61.

[15] Sarma, "On the Rationale of the Maxim Aṅkānāṃ Vāmato Gatiḥ," pp. 73–84, investigates the reasoning behind the right-to-left direction in reading the *bhūtasaṃkhyā.*

[16] Sarma, "On the Rationale of the Maxim Aṅkānāṃ Vāmato Gatiḥ," p. 65.

[17] Datta and Singh, *History of Hindu Mathematics,* p. 71.

[18] Chrisomalis, *Numerical Notation: A Comparative History,* p. 209.

[19] Plofker, Mathematics in India, p. 75.

[20] Annette Wilke and Oliver Moebius, *Sound and Communication: An Aesthetic Cultural History of Sanskrit Hinduism* (Berlin and New York: De Gruyter, 2011), p. 492. In relation to the tradition of Indian alphasyllabic numeration, Chrisomalis, *Numerical Notation: A Comparative History,* pp. 211–212, also mentions the *akṣarapallī* system used with great frequency in Jain manuscripts until the sixteenth century.

[21] The has been critically edited and translated in M. Raṅgācārya, ed., *The Ganita Sāra Saṃgraha of Mahāvīrācārya, with English Translation and Notes* (Madras: Government Press, 1912).

[22] See GSS 1.3–8.

[23] See Petrocchi, *The Gaṇita Sāra Saṃgraḥa.*

[24] See, for instance, the *Gaṇitatilaka* by Śrīpati (eleventh century C.E.).

[25] It is possible that this heading was added by an editor or scribe.

[26] See Raṅgācārya, *The Ganita Sāra Saṃgraha of Mahāvīrācārya,* p. 7.

[27] The printed edition reads *madri* here, instead of *mahīdra;* see Raṅgācārya, *The Ganita Sāra Saṃgraha of Mahāvīrācārya,* p. 7.

[28] The printed edition reads *pannagā* here, instead of *pannaga;* see Raṅgācārya, *The Ganita Sāra Saṃgraha of Mahāvīrācārya,* p. 7.

[29] Most of the last verses are unmetrical.

[30] The first part of this paper is partly the result of research I conducted for my 2012 MPhil thesis, *The Gaṇita Sāra Saṃgraḥa.* All the translations given here are my own, unless otherwise noted.

31 The correct reading should perhaps be *khecara* instead of *keśava,* which in no way is related to the number 9. On the other side, *khecara* means "planet," of which there are said to be nine in Hindu astrology and astronomy.

32 *Navaratna* and *navanidhi* are common expressions in Sanskrit literature. It is difficult to track the origin of the association of the number 9 to the term "jewel." However, I am inclined to think that this usage may be related to the number of the (nine) planets.

33 *Ṛgveda* 3.9.9 reads *trīṇi śatāni trisahasrāṇi triṃśa ca nava ca,* "three hundred and three thousand and thirty and nine."

34 See Petrocchi, *The Gaṇita Sāra Saṃgraḥa.*

35 Plofker, *Mathematics in India,* pp. 212–214, explains the role of the Sanskrit commentarial tradition of mathematics.

36 See Petrocchi, *The Gaṇita Sāra Saṃgraḥa.*

37 The term *udāharaṇa* is also found to denote "sample problem" in Sanskrit texts.

38 Plofker, *Mathematics in India,* pp. 173–178.38

39 I have unfortunately not been able to examine the manuscripts of the GSS. For his edition Raṅgācārya used five manuscripts, all of which are in Sanskrit. One is written in Grantha characters and contains the first five chapters of the work with a running commentary in Sanskrit. Two are palm-leaf manuscripts in Kannada characters, one of which contains the first five chapters and the other contains the seventh chapter on geometry. Another manuscript is a transcription on paper in Kannada characters of an original palm-leaf manuscript belonging to a Jaina pandit, which contains the whole of the work with a short commentary in Kannada. A fifth manuscript is a transcription on paper in Kannada characters of a palm-leaf manuscript found in a Jaina monastery in South Karnataka. It also contains the entire work and gives a brief statement of the problems and their answers in Kannada.

40 I find Raṅgācārya's translation, *The Ganita Sāra Saṃgraha of Mahāvīrācārya,* p. 8, of this verse imprecise:

> By means of the [following] eight qualities, viz., quick method in working, forethought as to whether a desirable result may arrived at, or as to whether an undesirable result will be produced, freedom from dullness, correct comprehension, power of retention, and the devising of new means in working, along with getting at those numbers which make (unknown) quantities known [by means of these quantities] as [an] arithmetician is to be known as such.

A similar translation is found in Plofker, *Mathematics in India,* p. 164, who translates *laghukaraṇa* with "swift work" and strangely renders *vyaktikarāṅkaviśiṣṭair* as "having the answers."

41 See for instance, *badhirīkṛ* ("to deafen"), *śuklīkṛ* ("whiten"), and *vājīkṛ* ("to strengthen").

42 It is interesting to notice that traditionally gaṇita, "mathematics, computation" was distinguished in vyaktagaṇita (lit., "manifest computation" or computation with known quantities) and avyaktagaṇita (lit., "unmanifest computation" or computation with unknown quantities).

43 The verb *vyaktīkṛ* literally means "to make manifest, clear."

44 In Sanskrit mathematical texts, *karaṇa* means "calculation, procedure."

45 *Ūha,* together with *tarka,* is a term belonging to the classical Indian tradition of logic and disputation. Both are linked to the relation between perception, logical reasoning, and means of knowledge.

46 *Apoha* is another terms belonging to classical Indian tradition of logic. The *apoha* theory was expounded in Dignāna and Dharmakirti's school, for instance.

47 *Arthaśāstra* 6.1.4 has "the qualities of the intellect" (*prajñāguṇāḥ*) of a king are *śuśrūṣāśravaṇagrahaṇadhāraṇavijñānohāpohatattvābhiniveśāḥ prajñāguṇāḥ.*

48 See Vettam Mani, *Purāṇic Encyclopaedia: A Comprehensive Dictionary with Special Reference to the Epic and Purāṇic Literature* (Delhi: Motilal Banarsidass, 1975, reprint), p. 62.

49 In *Jñānasiddhānta* 11.5, 11.11, and 11.18 eight supernatural faculties are mentioned. These parallel those found in *Paśupatasūtra* 1.23–26, 21–22, and 28–37. See Andrea Acri, Helen M. Creese, and Arlo Griffiths, eds., *From Laṅkā Eastwards: The Rāmayāṇa in the Literature and Visual Arts of Indonesia* (Leiden: KITLV Press, 2011), p. 80.

50 *Vargaṇa* refers to a sort of karmic molecules formed by a number of *varga*s, or karmic atoms.

51 See Raj Pruthi, *Jainism and Indian Civilization* (New Delhi: Discovery Publishing House, 2004), p. 45.

52 In his 1916 work *Course in General Linguistics* Ferdinand de Saussure details his views on the nature and structure of language. In his paradigm, a sign is the basic unit of communication and a mental construct. Signifier and signified are the two labels used to denote what constitute a sign. A signifier is the form that the sign takes, while the signified is the concept it represents.

53 See the list found in GSS 1.53–62, for which I provided a words-numbers correspondence above.

54 Donald Davidson, *Inquiries into Truth and Interpretation* (Oxford: Clarendon, 1984), p. xiii.

55 Sara Mills, *Discourse* (London: Routledge, 1997), p. 15.

56 I refer here to the practitioners of the *jyotiḥśāstra* branch of knowledge.

57 Staal, *Universals: Studies in Indian Logic and Linguistics,* p. 17.

58 Staal, *Universals: Studies in Indian Logic and Linguistics,* pp. 17, 130.

59 These four terms can be also defined as *lakṣya, bhūta, hetu,* and *lakṣaṇa.* I mean that a given number (*saṃkhyā*) is the *lakṣya,* i.e., it is the thing "to be defined"; an object is the *bhūta; hetu* is the "condition, reason"; and *lakṣaṇa* is the definendum, thus the technical term (*saṃjñā*). If a certain *bhūta* meets a certain condition (*hetu*), then it becomes a *lakṣaṇa* able to define a certain corresponding *lakṣya.* For instance, again in the case of *agni:* the *lakṣya* is the number 3, *bhūta* is the object "fire," *hetu* is the "three sacrificial fires," and the *lakṣaṇa* is the word *agni,* which has become a technical term standing for the number 3.

However, I prefer the formulation *saṃkhyā, bhūta, lakṣaṇa,* and *saṃjñā* because, in my opinion, it better clarifies the logic correspondence between the four elements.

[60] Staal, *Universals: Studies in Indian Logic and Linguistics,* pp. 130–131, uses this formula in investigating the concept of pakṣa in Indian thought.

[61] Staal, *Universals: Studies in Indian Logic and Linguistics,* p. 130, uses this kind of formula to analyze the notions of *pakṣa, hetu,* and *sādhya* in Indian traditional logic.

[62] Mills, *Discourse,* p. 26.

[63] Jonathan D. Culler, *Structuralist Poetics: Structuralism, Linguistics, and the Study of Literature* (London: Routledge and Kegan Paul, 1975), p. 36.

[64] See Roland Barthes, *Mythologies* (Paris: Éditions du Seuil, 1957).

[65] Genoveva Martí, "Reference," in Manuel García-Carpintero and Max Kölbel, eds., *The Continuum Companion to the Philosophy of Language* (London and New York: Continuum International Publishing Group, 2012), pp. 106–124.

[66] Pierre-Sylvain Filliozat, "Ancient Sanskrit Mathematics: An Oral Tradition and a Written Literature," in Karine Chemla, ed., *History of Science, History of Text* (Dordrecht, The Netherlands: Springer, 2004), pp. 137–157.

[67] Frits Staal, "Artificial Languages across Sciences and Civilizations," *Journal of Indian Philosophy* 34 (2006): 102.

[68] Staal, in "Artificial Languages across Sciences and Civilizations," gives the example of the mathematical notation used in the *Bakshālī Manuscript* (third to ninth centuries C.E.?).

[69] Staal, "Artificial Languages across Sciences and Civilizations," pp. 89–141.

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